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EXP1-3: A self-propelled vehicle

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1 Introduction

To make a self-propelled vehicle that go as close as to a target distance, we need to solve three major problems. Firstly, how to select energy storage devices and energy release principles. Secondly, how can we make sure the car go in a straight line. Lastly, ensure the car stop at the target distance. The important factors that need to be considered about including 3 parts. First of all, design cost, the lower the better. Next, sustainability. We are suggested to avoid using none recycling materials. Ultimately, vehicle weight, energy drive system, and wheel friction all have an impact on energy conversion efficiency. This will directly influence our car's performance.



Figure 1. Overall view of the vehicle

2 Design Strategy

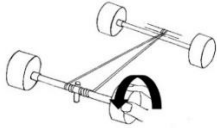
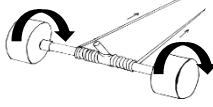
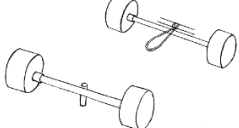
With the principle of "the less, the better", the self-propelled car is supposed to be of low-consumption, recyclability and high performance.

2.1 Power Driving System

A rubber-band-powdered device is designed as the energy resource of the rear-drive car. The energy system is primarily composed of the front axle, the rear axle and the rubber band.

In the process of driving, the rubber band is firstly wound manually backwards, transforming the human mechanical energy into the elastic potential energy of the rubber band. Then, the rubber band is released and, with the pull of the rubber band, the wheels rotate forward. In this case, the elastic potential energy converts into the kinetic energy of the car that enables it to move forward (Fig.1).

Table 1. Principle of car power module

Procedure	Loading (Annual Winding)	Releasing (Rubber Band Rebound)	Released (Car Moving)
Energy Transformation	Human Mechanical Energy ↓ Elastic Potential Energy of Rubber Bands	Elastic Potential Energy of Rubber Bands ↓ Kinetic Energy of Rear Axle	Kinetic Energy of Rear Axle ↓ Kinetic Energy of Car
Schemas			

2.2 Structural design and assembly

As shown in Fig.2, the car consists of six parts. To avoid energy loss due to accidental wheel idling, rubbers bands are applied as extra tread patterns to enhance the friction between car and the ground. Given the frames of back-rear car in current market, it is reasonable to adapt sides-paralleled design, which promotes stability and guarantees the car drives straight. With “#” bracket, the robustness is added and the material is saved.

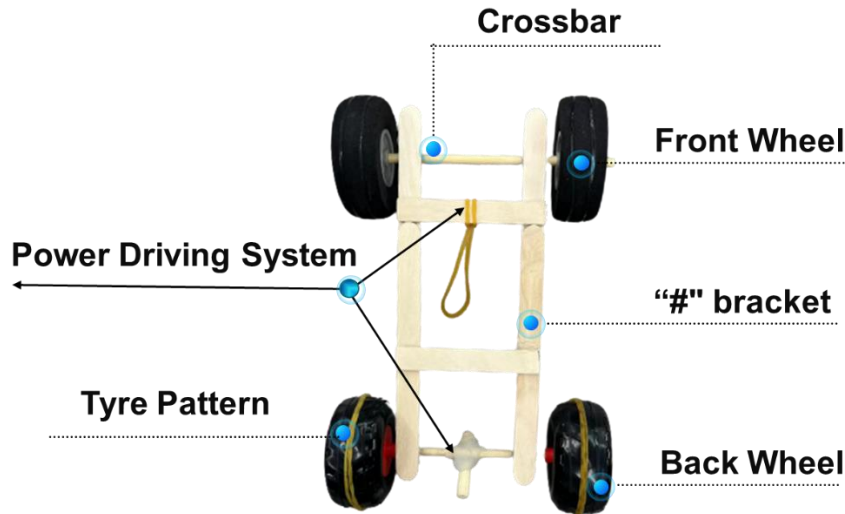


Figure 2. Bracket of the car

During the building procedure, ice cream sticks are initially used to make the “#” bracket, followed by tying the rubber band at the front crossbar and wrap extra bands around tyres to increase tread patterns. In order to fix the axles, ties are looped and connected on the bracket. Ultimately, cross the axle through the loops and connect wheels: the car is assembled (Fig. 3).

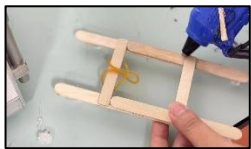
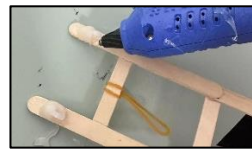
				
“#” bracket	Rubber-band Binding	Increase tyre tread pattern	Ties are looped & Secured to the bracket	Crossbar to car tyres

Figure 3. Process of car assembling

2.3 Analysis of pros and cons

Based on the universal analysis of the car, some advantages and disadvantages are listed below (Table 3).

Table 3. Advantages and disadvantages of design

Advantages	Disadvantages
✓ Low material cost (60 grams in total)	X Changeable elastic coefficient of rubber band by weather factor
✓ Good stability, ensuring straight driving	X Ground condition tends to affect moving car because of its lightness
✓ Increased tyre tread patterns and enough friction	X Dislocation of tread patterns, which is difficult to restore
✓ high energy sufficiency	
✓ Firm connection by hot melt glue	

2.4 Calculation

To calculate the energy conversion rate of the car, we combine the dynamics with the elastic potential energy of the rubber band.

Firstly, the stiffness k of the rubber band is obtained by curve fitting shown in Fig.4.

The stiffness of the rubber band is 39.433N/m .

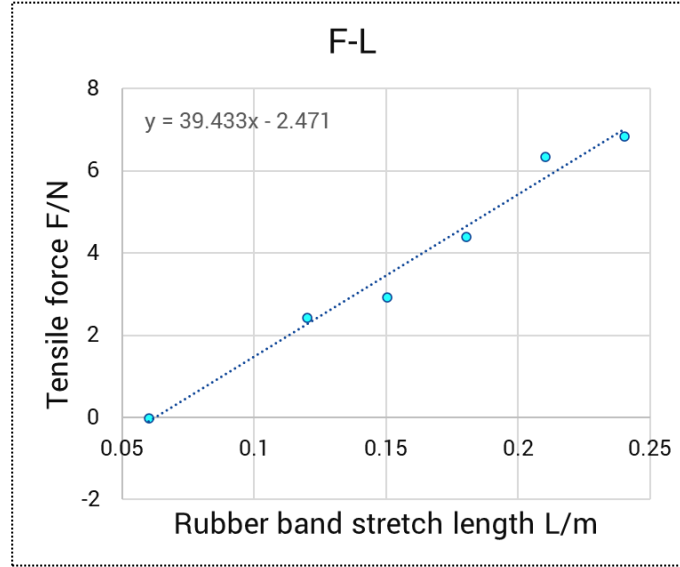


Figure 4. Rubber band stiffness curve $y=39.433x-2.471$

Subsequently, calculate the stretched length of the rubber band ΔL .

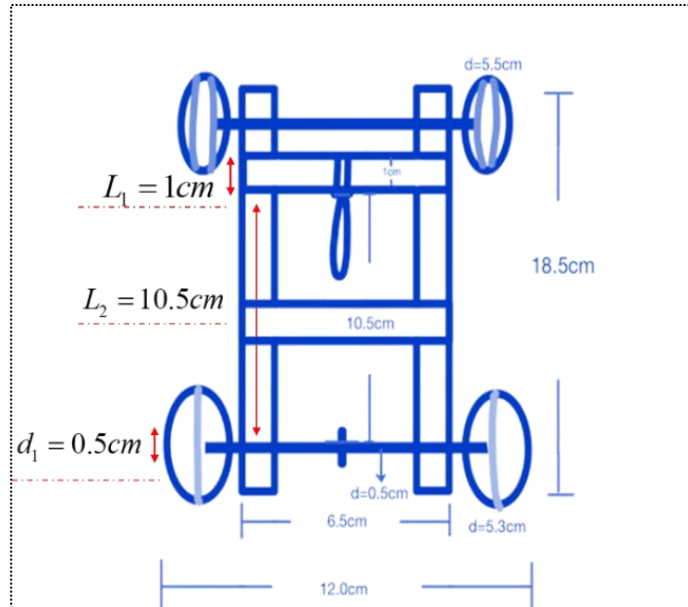


Figure 5. Overview of car

n is the number of rubber band wraps. The rubber band original length is $L_0=6\text{cm}$.

ΔL is obtained from the car overview shown in Fig.5 with formula (1).

$$\begin{aligned} \Delta L &= 2L_1 + L_2 + nL_3 - L_0 \\ L_3 &= \pi d_1 \end{aligned} \quad (1)$$

The stored energy E_{stored} is calculated with elastic potential energy formula (2). The equation of the relationship between E_{stored} and n is established.

$$E_{Stored} = \frac{1}{2}k(\Delta L)^2 \quad (2)$$

The kinetic energy E_d is calculated based on running distance and time with formula (3).

$$\begin{cases} S = 0.5at^2 \\ v_0 = at \\ E_d = 0.5mv^2 \end{cases} \quad (3)$$

Finally, the energy conversion rate of the car η is calculated with formula (4).

$$\eta = \frac{E_d}{E_{Stored}} \quad (4)$$

The relationship between the running distance S and the number of the rubber band warps n is established by fitting the data in prediction test. The result is shown in Fig.6.

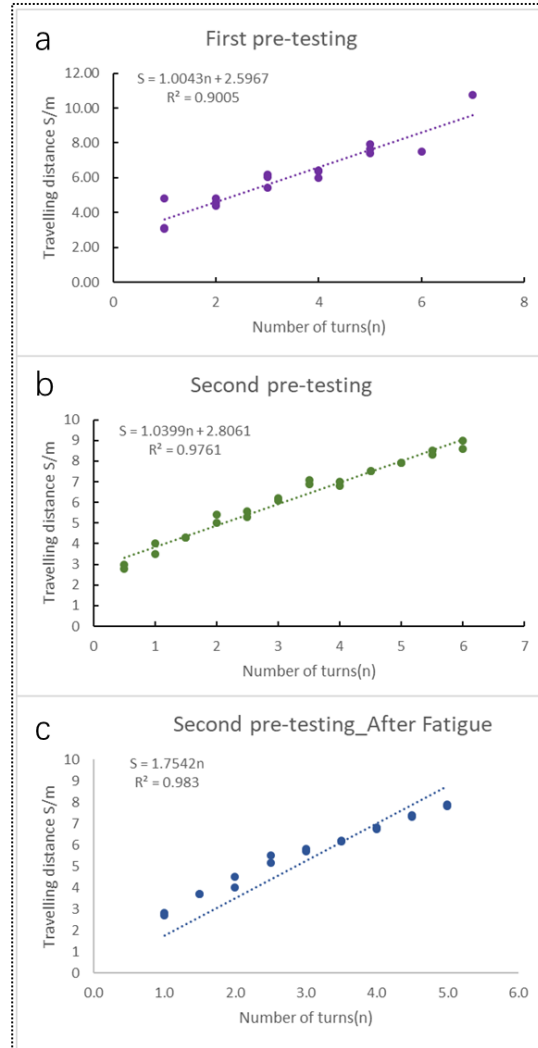


Figure. 6. S-n prediction curves for the car pretest

(a) represents the fitted curve $S = 1.0043n + 2.5967$ at the first pretest (b) is the fitted curve $S = 1.0399n + 2.8061$ at the second pretest (c) is the fitted curve after fatigue at the second pretest $S = 1.7542n$

The equation for the relationship between S and E_{stored} is calculated by combining the fitted S-n equations with equation (1)(2). Equations (5), (6), and (7) show the relationship between S and E_{stored} for the first pretest, the second pretest, and the second pretest after fatigue, respectively (k is the coefficient of strength).

$$S = \frac{200.86}{\pi} \sqrt{\frac{2E_{stored}}{k}} - \frac{13.0559}{\pi} + 2.5967 \quad (5)$$

$$S = \frac{207.98}{\pi} \sqrt{\frac{2E_{stored}}{k}} - \frac{13.5187}{\pi} + 2.8061 \quad (6)$$

$$S = \frac{350.84}{\pi} \sqrt{\frac{2E_{stored}}{k}} - \frac{22.8046}{\pi} \quad (7)$$

3 Design cost

In the process of selecting materials, six recyclable materials are chosen. Only rubber bands cost 0.6 QM\$. The total cost is -4.4QM\$.

Table 4. Material consumption chart

Material	Component Name	Specificities	Source
 Ice-cream Stick	"#" bracket	✓ Robustness ✓ Light ✓ Degradable	Leftovers from Zhang Yunyi's high school crafts
 Bamboo Chopsticks	Crossbar & Power Driving System	✓ Robustness ✓ Light ✓ Degradable	From the restaurants in Hoi Tin Court & Sing Tin Court
 Rubber Bands	Power Driving System & Tyre Pattern	✓ High Elasticity & Friction ✓ Recyclable	Bought at the QM shop.
 Model Wheels	Front and Back Wheels	✓ Light ✓ Stable ✓ Recyclable	Removed from Two model aeroplane
 Zip Ties	Connecting the Wheel to Bracket	✓ Light ✓ Solid ✓ Degradable	Left over from the club's model assembly
 Black Leather Band	Tyre Pattern	✓ Light ✓ High friction ✓ Recyclable	Small leather bands to tie your hair.

4 Sustainability

As shown in Fig.7 and Table 5, following the concept of environmental recycling, all the materials selected for the car are recyclable. Only Ice-cream Stick, Bamboo chopsticks, Zip Ties are non-reused. All reusable materials have been returned to the QMES shop.

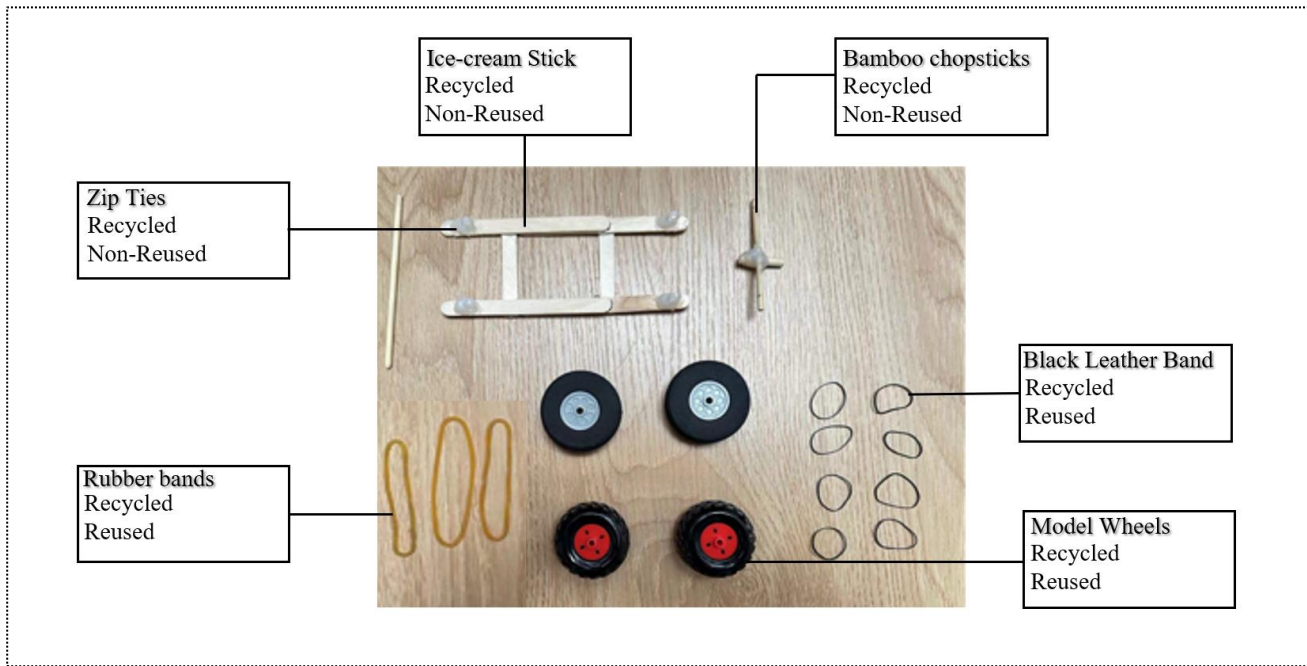


Figure 7. Vehicle parts recovery diagram

Table 5. Material recovery status

Material	Components	Reused or Not	Recycled or Not
Ice-cream Stick	"#" bracket	No	Yes
Bamboo chopsticks	Crossbar & Power Driving System	No	Yes
Rubber bands	Power Driving System & Tyre Pattern	Yes	Yes
Model Wheels	Front and Back Wheels	Yes	Yes
Zip Ties	Connecting the Wheel to Bracket	No	Yes
Black Leather Bands	Tyre Pattern	Yes	Yes

The energy required by the vehicle to travel is related to the distance travelled, as shown in formulas (5), (6) and (7). In addition, the elastic potential energy of the rubber band has been fully released when the car has just been started, and is converted into the mechanical energy of the car, the energy consumption is less. The subsequent process is to convert the mechanical energy of the car into friction consumption and internal energy.

With the principle of 'The less, the better.', the choice of all materials possesses a shared feature that they are all light in weight, leading to the overall weight of only 60g. In this case, the friction will be greatly reduced.

The extra tyre tread moderately adds some friction, so acceleration is not as great, while avoiding wheel spin, which reduces speed and air friction. The lower mass centre of the car and the '#' structure of the body ensures good stability when moving, the hollow car body had small contacting surface contributes to small air friction.

5 Prediction and Performance Evaluation

During the first and second tests, the body and material of the car were not modified, i.e., its cost was constant at -4.4QM\$ and its weight was constant at 60g.

Table 6. Prediction and Actual

	Prediction1	Race1	Prediction2	Race 2
Describe energy storage mechanism	Rubber band powered			
Total Weight Vehicle	0.06kg			
Number of rotations	4.0	4.0	4.5	4.5
Amount of energy stored	0.30025J	0.30025J	0.32248J	0.32248J
Energy conversion efficiency	26.80%	38.57%	30.31%	33.66%
Total Distance Travelled	6.38-6.80m	7.89m	7.6m	7.0m
Design Cost	-4.4	-4.4	-4.4	-4.4
Accuracy		17.76%		8.6%

In the first test, the car was expected to travel 6.7 meters with 4 winding loops, while it actually travelled 7.89 m. In the second test, the car was expected to travel 7.6 meters with 4.5 winding loops, while it actually travelled 7 m.

To estimate the prediction, the energy conversion rate bias δ and distance bias α were calculated using equations (8) and (9), respectively.

Energy conversion rate bias δ :

$$\delta = \frac{|\eta_{pre} - \eta_{actual}|}{\eta_{actual}} \quad (8)$$

Distance bias α :

$$\alpha = \frac{|S_{pre} - S_{actual}|}{S_{actual}} \quad (9)$$

The result is shown in Table 7.

Table 7. Calculation of errors

	Race 1	Race 2
Energy conversion rate bias δ :	30.52%	9.95%
Distance bias α	17.76%	8.6%

After the first racing, it is found that the actual data has great deviation from the prediction through calculating the deviation of the energy conversion rate and the distance bias. Based on the situation, analysis and improvements are made.

Problem existed:

- X The stiffness of the rubber band is greatly affected by the weather condition.
- X The rubber band will fatigue after times of testing, leading to the change of the stiffness so that affects the precision of the prediction.
- X As the car body is very light, so the change of distance per round is huge. The prediction based on rotation of integer rounds are not accurate enough.

Improvements:

- ✓ Substituting new rubber bands for the old one and conduct the pretesting again. Retest the relationship between the rotation rounds and distance after the band fatigues.
- ✓ Adding point five groups instead of testing only integer rounds.
- ✓ Testing multiple times at certain rounds to add up precision.

Table 8. The absolute coefficient of the fitted curve

	First Pretest	Second Pretest	Second Pretest After Fatigue
Absolute coefficient R^2	0.9005	0.9761	0.9830

After the modification, the fitted value is shown in Fig.6. The absolute coefficient R^2 gradually increases, representing the increase of the precision of the fitted curve (shown in Table 8.)

In addition, according to Table 7, it is found that the bias in the energy conversion rate is 9.95%, compared with 30.52% in the first time, and the bias in the distance change from 17.76% in the first time to 8.6%. Obvious improvements are vividly represented in the function of the car.

For the whole design, the shape similar with ‘#’ structure ensures stability of the car body, the zip ties perfectly connect the bracket and the axle and the added tyre pattern provides friction to avoid wheel idling. With the core of the design “The less, the better”, the common feature of all the materials is that they are both light in weight and degradable leads to a sustainable design and satisfies the requirements at the same time. In this design and modification, our car performed quite well.